

# **Assessed Problem Set 2**

## Machine Learning EFDS 2025

Group C

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# 1 Problem 1

## 1.1 Data Cleaning

### 1. Download files

See python file.

### 2.(a) Which variable measures smoking before pregnancy?

According to the natality data documentation (2013–2015), smoking before pregnancy is measured by the number of cigarettes the mother reports smoking in the three months prior to conception. This information is recorded in the variable `CIG0`, which captures the average number of cigarettes smoked per day before pregnancy.

**(b) Is mother’s race coded consistently across years?** No. Mother’s race is not coded consistently across years. During 2013–2015, some states used the older 1989 birth certificate, which records a single race category, while others used the 2003 revised certificate, which allows multiple races to be reported. Consequently, the raw race variable is not harmonised across states or years. The natality files therefore provide “bridged race” recodes (such as `MBRACE` and `MRACE31`), which must be used to obtain a consistent race measure across the three years.

### 3. How many observations do you end up with in each year?

After applying these restrictions, the cleaned sample sizes are:

2013: 39,251      2014: 37,285      2015: 30,965

### 4. Define treatment D

See python file

### 5. Summarize and visualise the data

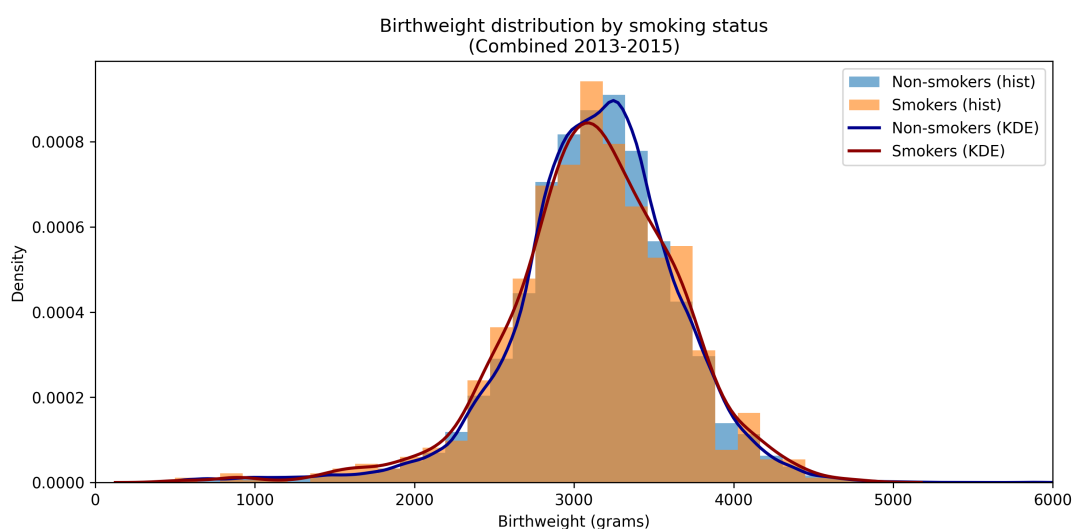


Figure 1: Birthweight distribution by maternal smoking status (combined 2013–2015). Density-scaled histograms and kernel density estimates (KDE) of infant birthweight (grams) for smoking and non-smoking mothers.

Figure 1 compares the distribution of infant birthweight for mothers who smoked before pregnancy and those who did not, using data pooled from 2013–2015. The histograms are plotted on a density scale to make the shapes of the two distributions directly comparable despite differences in sample sizes. Kernel density estimate (KDE) curves are added to provide smooth approximations of each distribution and to highlight differences that may be less visible in the histograms alone.

The two KDE curves differ in both their location and their overall shape. The non-smoker KDE peaks at a higher birthweight range, and its mass is more concentrated around this central region. In contrast, the smoker KDE peaks at a slightly lower birthweight range and is somewhat flatter, indicating a broader spread of birthweights within this group. The smoker curve also shows slightly more weight in the lower-birthweight region compared to the non-smoker curve, while the non-smoker curve places relatively more mass in the mid-to-upper birthweight range.

These differences highlight systematic shifts in the birthweight distribution between infants of smoking and non-smoking mothers.

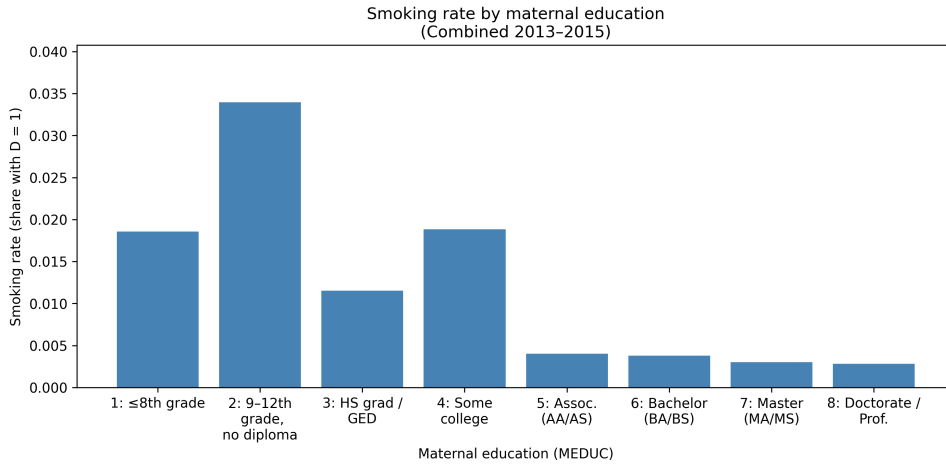


Figure 2: Smoking rate by maternal education (combined 2013–2015).

Figure 2 reports the *smoking rate* by maternal education, pooling all three years (2013–2015). For each education category  $MEDUC = m$  we compute

$$\text{Smoking rate}(m) = \mathbb{E}[D \mid MEDUC = m],$$

where  $D$  is the treatment indicator defined in part 1.4. By construction  $D = 1$  if the mother reports smoking any cigarettes in the three months before pregnancy and  $D = 0$  otherwise. Hence the smoking rate is simply the share of mothers with  $D = 1$  in each education group.

Using the share  $D = 1$  rather than raw counts is important because the number of births differs a lot across education categories. A bar chart in levels would mainly reflect how many mothers there are in each group, whereas the conditional mean  $\mathbb{E}[D \mid MEDUC = m]$  is directly interpretable as a probability of smoking for a mother with a given education. Pooling the years 2013–2015 further reduces noise and makes the pattern more stable.

The figure shows a clear negative relationship between education and smoking before pregnancy: smoking rates are highest for mothers with low education ( $MEDUC = 1$  or 2) and fall steadily as education increases. For the highest qualifications ( $MEDUC = 7$

and 8) the smoking rate is very close to zero, although these categories contain relatively few observations and the bars are therefore estimated less precisely. This graph is useful because it summarises, in a single plot, how an important risk factor (smoking) varies systematically with maternal socioeconomic status, which is likely to be a key confounder for birthweight in the subsequent analysis.

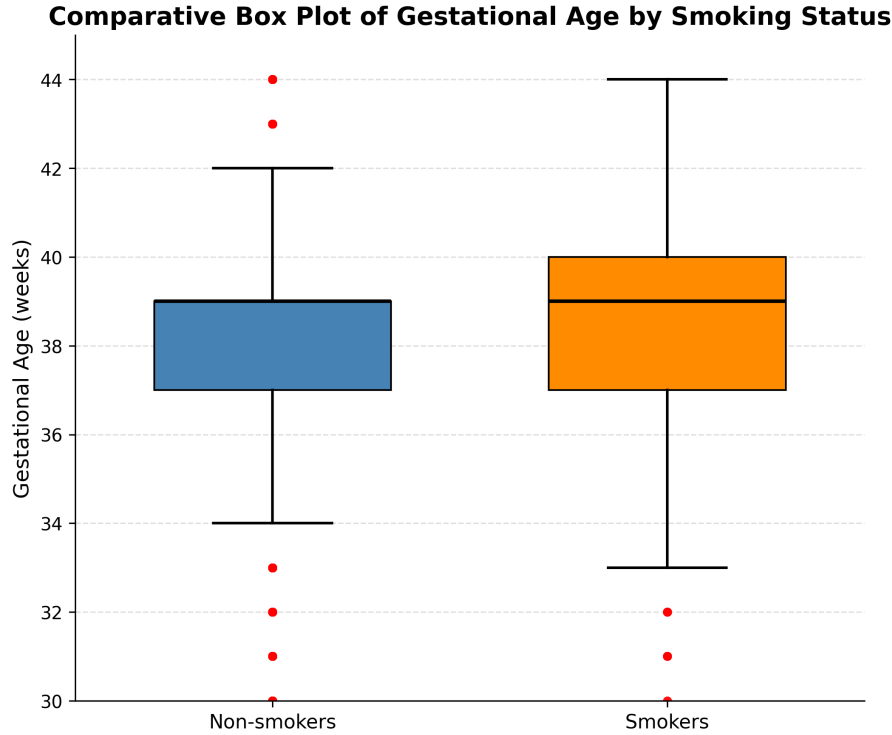


Figure 3: Comparative box plot of gestational age by smoking status (Combined 2013–2015). Distribution of gestational age for non-smokers and smokers. The  $y$ -axis is restricted to focus on the range where almost all births occur.

Figure 3 presents a comparative box plot of gestational age for smokers and non-smokers, using the combined 2013–2015 sample. Gestational age is an important descriptive variable because it is a strong determinant of birth outcomes and may differ systematically with maternal behaviour. For clarity, the  $y$ -axis is restricted to a narrower range (approximately 30–44 weeks) so that the quartiles and medians are easily visible. Outliers are indicated in red.

The figure indicates that both groups exhibit a tightly centred distribution around term (approximately 38–40 weeks). Smokers display slightly more variation and a marginally higher median gestational age, although the difference between groups is small in magnitude. The key takeaway is that median gestational age is similar across smoking status, but the distribution of gestational ages are much more widespread for smokers as compared to non-smokers.

## 1.2 Feature Construction

### 1. Creating dummies

See python file

## 2. Polynomial terms

See python file

## 3. Interact mother's age and education dummies

See python file

# 1.3 LASSO for Birthweight Prediction

## 1.

The objective of this section is to analyse the determinants of child birthweight using a LASSO regression model while ensuring that the effect of maternal smoking is not penalised. We use 5-fold cross-validation to select the tuning parameter  $\lambda$ , report the selected variables, and interpret the most important predictors. All relevant choices are documented below.

## 2. Methodology

**Data Preparation.** Covariates were standardised using `StandardScaler` to place all variables on a comparable scale. The smoking indicator was not standardised. The question does not ask for a double LASSO, something explored in 1.4. The library *sklearn* does not allow us to specifically choose smoking to be unpenalised. So to work around this, we applied a “scaling trick”: multiplying the smoking indicator by a large factor (1000) so that its effective penalty is negligible. The final LASSO design matrix included smoking (scaled) and all standardised covariates.

**Model.** We estimate the following model:

$$\text{birthweight}_i = \alpha \cdot \text{smoking}_i + X_i^\top \beta + \varepsilon_i,$$

where  $\alpha$  is unpenalised and  $\beta$  is penalised using the LASSO  $\ell_1$  norm. The tuning parameter  $\lambda$  is chosen via 5-fold cross-validation using `LassoCV`.

## 3. Results

The selected tuning parameter (to 6 decimal places) is:

$$\hat{\lambda} = 0.244228.$$

Figure 4 shows the selected variables and their effect on birthweight. Y axis titles are shortened so they can fit.

Figure 5 shows the 5-fold cross-validation mean squared error across the grid of candidate  $\lambda$  values. The chosen  $\lambda$  minimises the CV error.

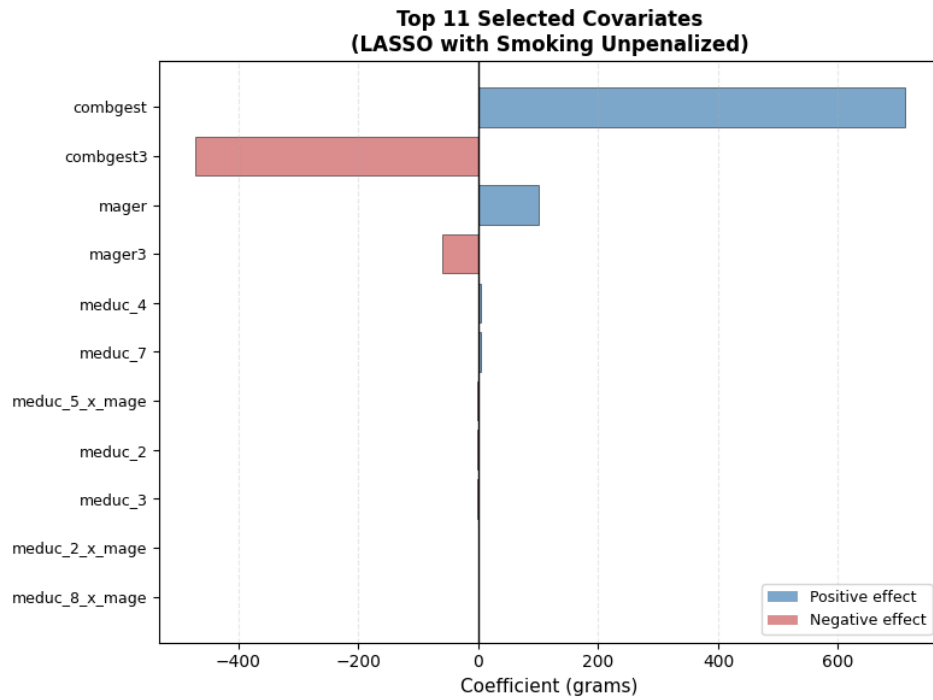


Figure 4: Selected Covariates with largest coefficients.

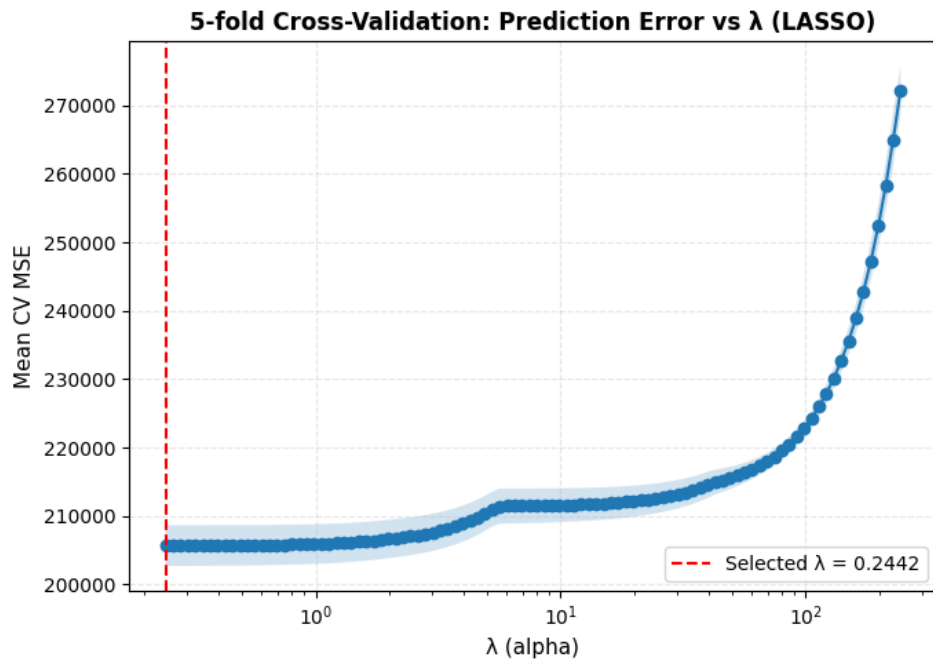


Figure 5: 5-fold CV: Prediction Error vs  $\lambda$ .

### 1.3.1 Interpretation

Maternal smoking has a large and negative effect on birthweight, estimated at approximately -30.3 grams.

Among the penalised covariates, the strongest predictors of higher birthweight include:

- **Gestational age**, which has a large positive effect and is medically known to be the most significant determinant of birthweight. This is represented on the graph by `combgest`.
- **Maternal age**, which is shown to have moderate positive relationship. This is represented by `mager`.
- **Maternal education**, which is shown to have a small positive relationship. This is shown by `meduc`.

### 1.3.2 Limitations

While the scaling trick ensures that the smoking indicator is effectively unpenalised within the LASSO optimisation, it introduces several limitations. First, the approach relies on choosing an arbitrarily large scaling factor, and the results can be sensitive to this choice: if the factor is too small, smoking is still partially penalised; if it is too large, numerical instability may occur and the optimisation path can change in unintended ways. Second, rescaling one variable alters the relative scale of the entire design matrix, which may lead the cross-validation routine to select a different tuning parameter  $\lambda$  than it would under correctly specified penalty factors. As a consequence, the set of selected covariates may change purely for numerical reasons rather than statistical ones. Finally, unlike methods that explicitly allow variable-specific penalty loadings, the scaling trick has no formal guarantee of reproducing the true “unpenalised” solution in finite samples. These issues suggest that this procedure is suboptimal.

## 1.4 Mechanical Double-LASSO Procedure

### 1. LASSO regression of $bw_i$ on $D_i$ and $X_i$

See python file

### 2. LASSO regression of $D_i$ on $X_i$

See python file

### 3. Union of selected controls

See python file

### 4. OLS regression

The estimated effect  $\hat{\tau} = -28.98$  and its standard error is 12.71

### 5. Interpretation

The estimated effect  $\hat{\tau}$  represents the causal effect of maternal smoking before pregnancy on child birthweight in grams, a proxy for child health. The result suggests that maternal smoking before pregnancy has an estimated average effect of reducing child birthweight by just under 29 grams. However, it should be noted that while the average is 29 grams, the effect is likely to depend on the number of cigarettes smoked, the more cigarettes smoked the larger the reduction in birthweight.

### 6. Visualisation

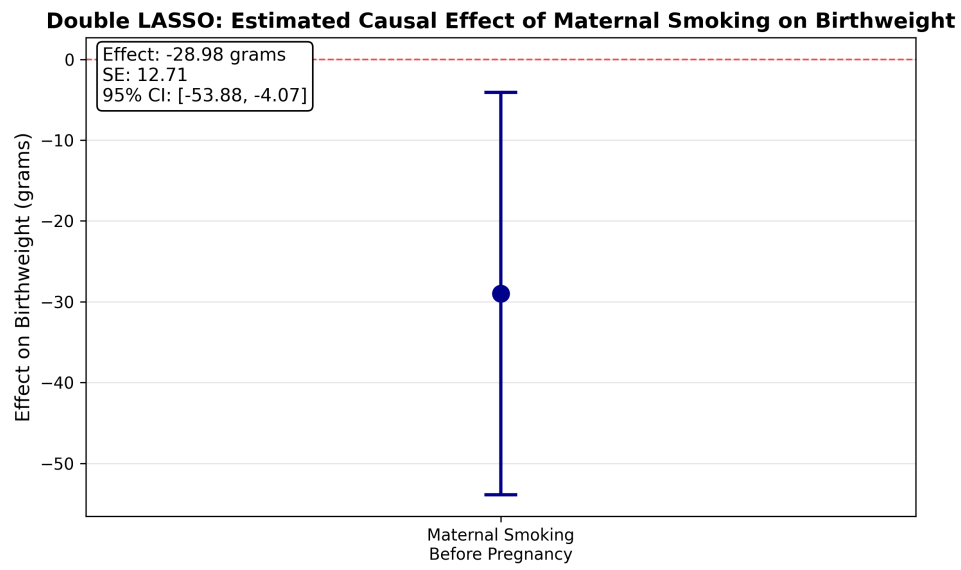


Figure 6: Estimated effect of maternal smoking on birthweight.

## Problem 2

### 2.1

(a) Let  $r = y - X\beta$ . Then (1) is equivalent to

$$\min_{r, \beta} \frac{1}{2} \|r\|_2^2 + \sum_{j=1}^p \lambda_j |\beta_j| \quad \text{s.t.} \quad r = y - X\beta.$$

(b) The constraint  $r = y - X\beta$  is in  $\mathbb{R}^n$ , so the Lagrange multiplier is  $u \in \mathbb{R}^n$ . The Lagrangian is

$$\mathcal{L}(r, \beta, u) = \frac{1}{2} \|r\|_2^2 + \sum_{j=1}^p \lambda_j |\beta_j| + u^\top (y - X\beta - r).$$

(c) For fixed  $(\beta, u)$ ,

$$\mathcal{L}(r, \beta, u) = \left( \frac{1}{2} \|r\|_2^2 - u^\top r \right) + u^\top (y - X\beta) + \sum_{j=1}^p \lambda_j |\beta_j|.$$

Only the bracket depends on  $r$ :

$$f(r) := \frac{1}{2} \|r\|_2^2 - u^\top r.$$

Then

$$\nabla_r f(r) = r - u = 0 \quad \Rightarrow \quad r = u,$$

and

$$f(u) = \frac{1}{2} \|u\|_2^2 - u^\top u = -\frac{1}{2} \|u\|_2^2.$$

Thus, after minimizing over  $r$  we get

$$\inf_r \mathcal{L}(r, \beta, u) = -\frac{1}{2} \|u\|_2^2 + u^\top (y - X\beta) + \sum_{j=1}^p \lambda_j |\beta_j|.$$

(d) Write  $X\beta = \sum_{j=1}^p x_j \beta_j$  where  $x_j$  is the  $j$ th column of  $X$ . Then

$$u^\top X\beta = \sum_{j=1}^p (x_j^\top u) \beta_j.$$

Define  $s_j := x_j^\top u$  and

$$\phi_j(\beta_j) := \lambda_j |\beta_j| - s_j \beta_j,$$

so that minimizing over  $\beta$  is equivalent to minimizing each  $\phi_j$  over  $\beta_j \in \mathbb{R}$ .

For  $\beta_j > 0$ ,

$$\phi_j(\beta_j) = (\lambda_j - s_j) \beta_j;$$

for  $\beta_j < 0$ ,

$$\phi_j(\beta_j) = (-\lambda_j - s_j) \beta_j.$$

If  $|s_j| \leq \lambda_j$ : Then  $\lambda_j - s_j \geq 0$  and  $-\lambda_j - s_j \leq 0$ . Hence  $\phi_j$  decreases towards 0 from both sides and is minimized at

$$\beta_j = 0, \quad \phi_j(0) = 0,$$

so

$$\inf_{\beta_j \in \mathbb{R}} \phi_j(\beta_j) = 0 \quad \text{if } |s_j| \leq \lambda_j.$$

If  $|s_j| > \lambda_j$ : If  $s_j > \lambda_j$ , then on  $(0, \infty)$  we have  $\phi_j(\beta_j) = (\lambda_j - s_j)\beta_j \rightarrow -\infty$  as  $\beta_j \rightarrow +\infty$ . If  $s_j < -\lambda_j$ , then on  $(-\infty, 0)$  we have  $\phi_j(\beta_j) = (-\lambda_j - s_j)\beta_j \rightarrow -\infty$  as  $\beta_j \rightarrow -\infty$ . Thus

$$\inf_{\beta_j \in \mathbb{R}} \phi_j(\beta_j) = -\infty \quad \text{if } |s_j| > \lambda_j.$$

In summary,

$$\inf_{\beta_j \in \mathbb{R}} \phi_j(\beta_j) = \begin{cases} 0, & |s_j| \leq \lambda_j, \\ -\infty, & |s_j| > \lambda_j. \end{cases}$$

Equivalently, the infimum over all  $\beta$  is finite if and only if

$$|x_j^\top u| = |s_j| \leq \lambda_j \quad \text{for all } j.$$

(e) The dual function is

$$g(u) := \inf_{r, \beta} \mathcal{L}(r, \beta, u).$$

From (c) and (d),

$$g(u) = \begin{cases} -\frac{1}{2}\|u\|_2^2 + u^\top y, & \text{if } |x_j^\top u| \leq \lambda_j \quad \forall j, \\ -\infty, & \text{otherwise.} \end{cases}$$

Thus the dual problem is

$$\max_{u \in \mathbb{R}^n} \left( u^\top y - \frac{1}{2}\|u\|_2^2 \right) \quad \text{s.t.} \quad |x_j^\top u| \leq \lambda_j, \quad j = 1, \dots, p,$$

with dual-feasibility condition

$$|x_j^\top u| \leq \lambda_j \quad \text{for all } j = 1, \dots, p.$$

## 2.2

(a) From the Lagrangian in part 1(b),

$$\mathcal{L}(r, \beta, u) = \frac{1}{2}\|r\|^2 + \sum_{j=1}^p \lambda_j |\beta_j| + u^\top (y - X\beta - r),$$

the first order condition with respect to  $r$  is

$$\nabla_r \mathcal{L} = r - u = 0 \quad \Rightarrow \quad r^* = u^*.$$

(b) Collecting the  $\beta_j$  terms of the Lagrangian:

$$L(\beta, \cdot, u) = \sum_{j=1}^p (\lambda_j |\beta_j| - \beta_j x_j^\top u) + (\text{terms independent of } \beta).$$

The subgradient optimality condition is

$$0 \in -x_j^\top u^\star + \lambda_j \partial |\beta_j^\star|.$$

Using  $r^\star = u^\star$ ,

$$x_j^\top r^\star \in \lambda_j \partial |\beta_j^\star|.$$

Since

$$\partial |t| = \begin{cases} \{\text{sign}(t)\}, & t \neq 0, \\ [-1, 1], & t = 0, \end{cases}$$

we obtain:

$$\begin{aligned} \beta_j^\star = 0 &\Rightarrow |x_j^\top r^\star| \leq \lambda_j, \\ \beta_j^\star \neq 0 &\Rightarrow x_j^\top r^\star = \lambda_j \text{sign}(\beta_j^\star). \end{aligned}$$

(c) Assume we know a dual feasible point  $\hat{u}$  and  $R > 0$  such that the optimal dual solution lies in

$$B = \{u : \|u - \hat{u}\|_2 \leq R\}.$$

For any  $u \in B$ ,

$$|x_j^\top u| = |x_j^\top \hat{u} + x_j^\top (u - \hat{u})| \leq |x_j^\top \hat{u}| + |x_j^\top (u - \hat{u})|.$$

By Cauchy-Schwarz,

$$|x_j^\top (u - \hat{u})| \leq \|x_j\|_2 \|u - \hat{u}\|_2 \leq R \|x_j\|_2.$$

Therefore,

$$\sup_{u \in B} |x_j^\top u| \leq |x_j^\top \hat{u}| + R \|x_j\|_2.$$

(d) If for some coordinate  $j$ ,

$$|x_j^\top \hat{u}| + R \|x_j\|_2 < \lambda_j,$$

then for all  $u \in B$ ,

$$|x_j^\top u| < \lambda_j.$$

for  $u^\star \in B$ ,

$$|x_j^\top r^\star| = |x_j^\top u^\star| < \lambda_j.$$

From part (b),

$$\beta_j^\star = 0.$$

## 2.3

For the screening procedure to be effective, the choice of the approximate dual point  $\hat{u}$  and the radius  $R$  must meet several criteria.

First,  $\hat{u}$  is typically chosen to be dual feasible, and  $R$  should be chosen so that we can guarantee  $u^* \in B$  (for example, via a bound based on the duality gap). Without feasibility, no correctness guarantees can be established for the screening rule.

Second,  $\hat{u}$  should closely approximate  $u^*$ , since the accuracy of this approximation determines the magnitude of  $R$ . A bad approximation forces  $R$  to be larger to maintain the guarantee that  $u^* \in B$ , and a larger radius weakens the screening bound, which then reduces the number of features that can be safely eliminated. Therefore, the radius  $R$  should be minimised so that the resulting bound on  $\sup_{u \in B} |x_j^\top u|$  remains tight.

Third, the computation of both  $\hat{u}$  and  $R$  should be efficient, relying on quantities that the optimisation algorithm already produces, like the residual or the duality gap, thus avoiding any substantial computational overhead.

Finally, an effective choice of  $(\hat{u}, R)$  should improve progressively during optimisation. As the algorithm approaches optimality,  $\hat{u}$  should converge toward the true dual solution and  $R$  should contract accordingly. This principle is the foundation for the design of modern safe screening frameworks, including the SAFE rules (El Ghaoui, Viallon & Rabbani, 2012) and the Gap Safe rules (Fercoq, Gramfort & Salmon, 2015), which rely on increasingly precise dual approximations to guarantee correct and efficient exclusion of inactive features.

## 2.4

When using cyclic coordinate descent (CCD) to solve the LASSO, the safe screening rule helps speed up the algorithm by identifying coefficients that never need to be updated. The screening condition derived from the dual ball, that a feature  $j$  satisfies  $|x_j^\top \hat{u}| + R\|x_j\|_2 < \lambda_j$ , ensures that the corresponding coefficient  $\beta_j^*$  is zero at optimality. When this certificate is available, the CCD can leave out coordinate  $j$ , shrinking the set of active variables, and avoiding updates to coefficients that will not change from zero.

As the algorithm progresses and both the dual approximation  $\hat{u}$  and radius  $R$  are refined (for example, through reductions in the duality gap), the screening region tightens, allowing more features to be safely discarded. This results in a decreasing effective problem dimension, improving convergence speed and per iteration efficiency, whilst maintaining exactness of the solution. Therefore, safe screening turns CCD into a more focused algorithm that spends time on features that might matter.

## References

- [1] Laurent El Ghaoui, Vivian Viallon, and Tarek Rabbani. *Safe Feature Elimination in Sparse Supervised Learning*. *Pacific Journal of Optimization*, 8(4):667–698, 2012.
- [2] Olivier Fercoq, Alexandre Gramfort, and Joseph Salmon. *Mind the Duality Gap: Safer Rules for the Lasso*. In *ICML*, pages 333–342, 2015.